

IMO PHASE III LEVEL 2

BINOMIAL COEFFICIENTS

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1 Lucas' Theorem and Kummer's Theorem

What is the remainder of $\binom{m}{n} \pmod{p}$? And if $\binom{m}{n}$ is divisible by p , what is the p -adic valuation? These two questions are answered by **Lucas' Theorem** and **Kummer's Theorem** respectively.

Theorem 1.1 (Lucas' Theorem). Write the numbers n and k in base p : $n = \overline{n_d n_{d-1} \cdots n_0}_p$, $k = \overline{k_d k_{d-1} \cdots k_0}_p$. Then $\binom{n}{k} \equiv \prod_{i=0}^d \binom{n_i}{k_i} \pmod{p}$, with $\binom{n}{k} = 0$ when $n < k$. In particular, $\binom{n}{k}$ is divisible by p iff at least one of the base- p digits of n is greater than the corresponding base- p digit of k .

Lemma 1.2. For p prime and $x, r \in \mathbb{Z}$, $(1+x)^{p^r} \equiv 1 + x^{p^r} \pmod{p}$.

Proof. For all $1 \leq k \leq p-1$, $\binom{p}{k} \equiv 0 \pmod{p}$. Then we have

$$\begin{aligned} (1+x)^p &\equiv \binom{p}{0} + \binom{p}{1}x + \binom{p}{2}x^2 + \cdots + \binom{p}{p-1}x^{p-1} + \binom{p}{p}x^p \\ &\equiv 1 + x^p \pmod{p} \end{aligned}$$

Assume we have $(1+x)^{p^k} \equiv 1 + x^{p^k} \pmod{p}$. Then

$$\begin{aligned} (1+x)^{p^{k+1}} &\equiv \left((1+x)^{p^k} \right)^p \equiv \left(1 + x^{p^k} \right)^p \\ &\equiv 1 + x^{p^{k+1}} \pmod{p} \end{aligned}$$

□

Example 1.3 (Lucas). Suppose $p = 5, n = 482 = \overline{3412}_5, k = 176 = \overline{1201}_5$. By the lemma we have

$$\begin{aligned} (1+x)^{482} &\equiv (1+x)^{125(3)} \cdot (1+x)^{25(4)} \cdot (1+x)^{5(1)} + (1+x)^2 \pmod{5} \\ &\equiv (1+x^{125})^3 \cdot (1+x^{25})^4 \cdot (1+x^5)^1 \cdot (1+x)^2. \end{aligned}$$

Since the only way to get x^{176} is by three x^{125} terms, two x^{25} terms, zero x^5 terms, and one x^1 terms on the

RHS. Hence the number of ways to get such term is

$$\binom{482}{176} \equiv \binom{3}{1} \binom{4}{2} \binom{1}{0} \binom{2}{1}.$$

Proof. (Lucas') The proof of Lucas' theorem follow a similar structure of the example. Another way to proof the theorem is by writting $n = Np + i, k = Kp + j$, and proceed by MI with some pascal's triangle properties. $\square$

Example 1.4 (lucas). *Prove that the number of odd binomial coefficients in n -th row of Pascal triangle is equal to 2^r , where r is the number of 1's in the binary expansion of n .*

Proof. Write $n = \overline{n_d n_{d-1} \cdots n_0}$, $\binom{n}{k}$ is odd iff $k_i \leq n_i \forall i$, and hence there are 2 choices for k_i if $n_i = 1$ and 1 choice otherwise, i.e. the number of odd entries equal 2^r . $\square$

Definition 1.5 (p -adic valuation). *Given a non-zero integer n and a prime p , we define the p -adic valuation of n as the highest power of p that divides n , and write it as $v_p(n)$.*

Theorem 1.6 (Kummer's Theorem). *Given a prime p and integers m, n , $v_p\left(\binom{m}{n}\right)$ equals the number of carries in adding $m - n$ and n in base p .*

Theorem 1.7 (Legendre's Formula). *Let $n \in \mathbb{N}$, p be prime. Let $s_p(n)$ be the sum of the digits of n in base p . Then*

$$v_p(n!) = \frac{n - s_p(n)}{p - 1}.$$

Proof. Let $n = \overline{a_k a_{k-1} \cdots a_0}_p$ be base p representation. We have $s_p(n) = \sum_{i=0}^k a_i$, then

$$\begin{aligned} v_p(n!) &= \left\lfloor \frac{n}{p} \right\rfloor + \left\lfloor \frac{n}{p^2} \right\rfloor + \left\lfloor \frac{n}{p^3} \right\rfloor + \cdots \\ &= (a_k p^{k-1} + \cdots + a_1) + (a_k p^{k-2} + a_{k-1} p^{k-3} + \cdots + a_2) + \cdots + (a_k p + a_{k-1}) + a_k \\ &= a_k \frac{p^k - 1}{p - 1} + a_{k-1} \frac{p^{k-1} - 1}{p - 1} + \cdots + a_1 \\ &= \frac{1}{p - 1} ((a_k p^k + a_{k-1} p^{k-1} + \cdots + a_0) - (a_k + a_{k-1} + \cdots + a_0)) \\ &= \frac{n - s_p(n)}{p - 1}. \end{aligned} \quad \square$$

Proof. (Kummer's) Write $m = \overline{a_r a_{r-1} \cdots a_0}_p, m - n = \overline{b_r b_{r-1} \cdots b_0}_p, n = \overline{c_r c_{r-1} \cdots c_0}_p$. Define carry func γ :

$$\gamma_i = \begin{cases} 1 & \text{if } b_i + c_i + \gamma_{i-1} \geq p, \\ 0 & \text{if } b_i + c_i + \gamma_{i-1} < p. \end{cases} \quad \forall 0 \leq i < r, \gamma_{-1} = 0.$$

Since a_r is the leading coeff of m , $\gamma_r = 0$. And observe that $a_i = b_i + c_i + \gamma_{i-1} - p\gamma_i \forall 0 \leq i \leq r$. By Legendre,

$$\begin{aligned} v_p\left(\binom{m}{n}\right) &= v_p(m!) - v_p((m - n)!) - v_p(n!) \\ &= \frac{m - s_p(m)}{p - 1} - \frac{m - n - s_p(m - n)}{p - 1} - \frac{n - s_p(n)}{p - 1} \\ &= \frac{\sum_{i=0}^r b_i + c_i - a_i}{p - 1} \\ &= \frac{p\gamma_0 + (p\gamma_1 - \gamma_0) + \cdots + (p\gamma_{r-1} - \gamma_{r-2}) - \gamma_{r-1}}{p - 1} \\ &= \gamma_0 + \gamma_1 + \cdots + \gamma_{r-1}. \end{aligned} \quad \square$$

2 Some other Theorem

Theorem 2.1 (Wilson's Theorem). For any prime p (and for primes only), $(p-1)! \equiv -1 \pmod{p}$.

Theorem 2.2 (Wolstenholme's Theorem). If $p \geq 5$, then $\binom{2p}{p} \equiv 2 \pmod{p^3}$, or equivalently, $\binom{2p-1}{p-1} \equiv 1 \pmod{p^3}$.

Proof. We have $\binom{2n}{n} = \sum_{k=0}^n \binom{n}{k}^2$. Since $\binom{p}{0} = \binom{p}{p} = 1$, we need to prove $\sum_{k=1}^{p-1} \binom{p}{k}^2 \equiv 0 \pmod{p^3}$. Since all the binomial coefficients are divisible by p , we can rewrite it as

$$\sum_{k=1}^{p-1} \left(\frac{1}{p} \binom{p}{k} \right)^2 \equiv 0 \pmod{p}.$$

But we have

$$\frac{1}{p} \binom{p}{k} = \frac{(p-1) \cdots (p-(k-1))}{k(k-1) \cdots 1} \equiv \frac{1(-1) \cdots (-(k-1))}{k(k-1) \cdots 1} \equiv \pm \frac{1}{k} \pmod{p},$$

thus

$$\sum_{k=1}^{p-1} \left(\frac{1}{p} \binom{p}{k} \right)^2 \equiv \sum_{k=1}^{p-1} \left(\pm \frac{1}{k} \right)^2 \equiv \sum_{k=1}^{p-1} \left(\frac{1}{k} \right)^2 = \sum_{k=1}^{p-1} k^2 = \frac{p(p-1)(2p-1)}{6} \equiv 0 \pmod{p}$$

as when $1/k$ go through 1 to $p-1$, so does k . □

Remark. An equivalent formulation is the congruence $\binom{ap}{bp} \equiv \binom{a}{b} \pmod{p^3}$ for $p \geq 5$, which we will not cover the proof here.

3 Problems for Discussion

Problem 1. For prime p , prove that $\binom{p}{n} - \lfloor \frac{n}{p} \rfloor$ is divisible by p .

Problem 2. Prove that

- All the binomial coefficients $\binom{n}{k}$, where $0 < k < n$, are divisible by p iff n is a power of p .
- All the binomial coefficients $\binom{n}{k}$, where $0 \leq k \leq n$ are not divisible by p iff $n+1$ is divisible by p^d , in other words, all the digits of n , except the leftmost, in base p are equal to $p-1$.

Problem 3. (VN TST 2010 P6) Let S_n be sum of squares of the coefficient of the polynomial $(1+x)^n$. Prove that $S_{2n} + 1$ is not divisible by 3.

Problem 4. Prove that $\text{lcm}(1, 2, \dots, n) = \text{lcm}(\binom{n}{0}, \binom{n}{1}, \dots, \binom{n}{n})$ iff $n+1$ is prime.

Problem 5. (Olimpiada Rioplatense 2015 Level 3 P5) For a positive integer number n we denote $d(n)$ as the greatest common divisor of the binomial coefficients $\binom{n+1}{n}, \binom{n+2}{n}, \dots, \binom{2n}{n}$. Find all possible values of $d(n)$

Problem 6. Prove that if we consider all the elements of the two sets

$$\left\{ \binom{2^n-1}{1}, \binom{2^n-1}{3}, \binom{2^n-1}{5}, \dots, \binom{2^n-1}{2^n-1} \right\} \quad \text{and} \quad \{1, 3, 5, \dots, 2^n-1\}$$

as a reminders modulo 2^n , then these sets coincide.

Problem 7. (Iran TST 2012 -first day- problem 1) Find all positive integers $n \geq 2$ such that for all integers i, j that $0 \leq i, j \leq n$, $i+j$ and $\binom{n}{i} + \binom{n}{j}$ have same parity.

Problem 8. Show that, for every prime number p and all integer r , there is a non negative integer n , such that:

$$\binom{2n}{n} \equiv r \pmod{p}$$

4 Problems (Combinatorial)

Problem 1. (2019 Pan-African Mathematics Olympiad P6) Find the 2019th strictly positive integer n such that $\binom{2n}{n}$ is not divisible by 5.

Problem 2. Wanda the Worm likes to eat Pascal's triangle. One day, she starts at the top of the triangle and eats $\binom{0}{0} = 1$. Each move, she travels to an adjacent positive integer and eats it, but she can never return to a spot that she has previously eaten. If Wanda can never eat numbers a, b, c such that $a + b = c$, prove that it is possible for her to eat 100,000 numbers in the first 2011 rows given that she is not restricted to traveling only in the first 2011 rows.

(Here, the $n + 1$ st row of Pascal's triangle consists of entries of the form $\binom{n}{k}$ for integers $0 \leq k \leq n$. Thus, the entry $\binom{n}{k}$ is considered adjacent to the entries $\binom{n-1}{k-1}, \binom{n-1}{k}, \binom{n-1}{k+1}, \binom{n}{k-1}, \binom{n}{k}, \binom{n}{k+1}$.)

Problem 3. $n(n \geq 3)$ integers $a_1, a_2, \dots, a_n$ lie around a circle in that order. In every operation, we calculate the absolute value of difference between each pair of two adjacent numbers to get n new integers, and then set them around the circle without disorganizing the order. Determine all the positive integer n , such that for any n integers, we can take finite operations to make all n integers are the same eventually.

Problem 4. (2019 China TST Test 4 P6) Given positive integer n, k such that $2 \leq n < 2^k$. Prove that there exist a subset A of $\{0, 1, \dots, n\}$ such that for any $x \neq y \in A$, $\binom{y}{x}$ is even, and

$$|A| \geq \frac{\binom{k}{\lfloor \frac{k}{2} \rfloor}}{2^k} \cdot (n + 1)$$

5 Problems (Functional Equation)

Problem 1. Determine all functions f defined on the set of all positive integers and taking non-negative integer values, satisfying the three conditions:

- (i) $f(n) \neq 0$ for at least one n ;
- (ii) $f(xy) = f(x) + f(y)$ for every positive integers x and y ;
- (iii) there are infinitely many positive integers n such that $f(k) = f(n - k)$ for all $k < n$.

Problem 2. Show that a sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ of plus minus one is periodic of period a power of 2 if and only if $\varepsilon_n = (-1)^{P(n)}$ where $P(n)$ is a polynomial with rational coefficients which is integer-valued.

Problem 3. (2009 USA TST P3) For each positive integer n , let $c(n)$ be the largest real number such that

$$c(n) \leq \left| \frac{f(a) - f(b)}{a - b} \right|$$

for all triples (f, a, b) such that

- f is a polynomial of degree n taking integers to integers, and
- a, b are integers with $f(a) \neq f(b)$.

Find $c(n)$.

Problem 4. (2016 Taiwan 1st TST IMO Mock P3) Let $\mathbb{Z}^+$ denote the set of all positive integers. Find all surjective functions $f : \mathbb{Z}^+ \times \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ that satisfy all of the following conditions: for all $a, b, c \in \mathbb{Z}^+$,

- $f(a, b) \leq a + b$;
- $f(a, f(b, c)) = f(f(a, b), c)$; and
- Both $\binom{f(a, b)}{a}$ and $\binom{f(a, b)}{b}$ are odd numbers. (where $\binom{n}{k}$ denotes the binomial coefficients)

Problem 5. (RMM2011, P2, Day 1) Determine all positive integers n for which there exists a polynomial $f(x)$ with real coefficients, with the following properties:

- for each integer k , the number $f(k)$ is an integer if and only if k is not divisible by n ;
- the degree of f is less than n .